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CASE REPORTS

Survival After Bilateral Adrenal Hemorrhage During Heparin Therapy

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BECAUSE it is a condition which continues to be associated with a high mortality rate, despite the fact that treatment is available for it, spontaneous adrenal hemorrhage deserves greater clinical attention. Many factors are said to predispose to such hemorrhage, including septicemia, pregnancy, hemorrhagic diseases, trauma, leukemia, lymphoma and other neoplasia.[30, 31] Anticoagulant therapy has been considered causative in some 35 patients described in the literature, only three of whom survived.[1-4] Both heparin and coumarin compounds have been implicated. The present case report describes a patient receiving heparin therapy who survived bilateral adrenal hemorrhage. It illustrates some of the difficulties of making an early diagnosis in this condition. The pathophysiology of this disorder is briefly reviewed.

A 74-year-old veteran first presented at the Queen Mary Veterans Hospital in 1949 with a possible myocardial infarction. He was hospitalized in 1951 for mild heart failure which responded to digitalization. On January 14, 1966, he was admitted again because of moderate dyspnea on exertion, orthopnea, intermittent paroxysmal nocturnal dyspnea, and severe cough especially when prone, for many years. He sought medical advice because of the increasing severity of his symptoms. He had smoked two packages of cigarettes daily until 1944 but none since. His sputum had recently become purulent and amounted to one cup per day. No fever or chills had been noted. It could not be determined definitely whether he had angina pectoris. Shortly before admission he had pain in his right calf. Intermittent ankle edema and bilateral varicose veins had been present for many years. There was no history of exposure to tuberculosis.

On examination he was afebrile and had an irregular heart rate of 156 per minute, a blood pressure of 110/75 mm. Hg in both arms and no jugular venous distension. Coarse rhonchi were present throughout both lungs, but no rales. His heart sounds were normal and there were no murmurs or friction rubs. All pulses except the right dorsalis pedis and both posterior tibials were present, and no bruits were detected. A non-tender thrombosed vein was noted on the medial aspect of the right calf. Neurological examination was negative except for absent ankle jerks. No excessive pigmentation was noted. A radiograph of the chest was negative and the electrocardiogram showed atrial fibrillation with a fast ventricular response. The patient was digitalized, intermittent positive-pressure breathing was started and, because of a moderate growth of pneumococci from his sputum, he was given penicillin. On this regimen his symptomatic and clinical improvement was marked.

On January 18, signs of a superficial thrombophlebitis appeared in the right calf. He was treated with bed rest, hot foments, and elevation of the leg. In addition, heparin 50 to 75 mg. was administered intravenously every six hours to keep the clotting time four hours after the injection at 25 to 30 minutes (control: 7 minutes). Despite these measures, over the next five to six days the inflammation spread up to the saphenofemoral junction.

On January 25 the patient had an episode of vague but severe pain, chiefly in the anterior chest but also in either arm or neck, accompanied by nausea and vomiting, lasting one-half to one hour, and requiring meperidine (Demerol) for relief. These episodes recurred intermittently over the next 48 hours. His blood pressure, which previously had been 110-140/70-90 mm. Hg (average 120/80), increased to 120-150/60-90 mm. Hg (average: 130/75). The electrocardiogram showed changes suggestive of a posterolateral infarction. Serum enzymes were elevated and subsequently reached a peak (SGOT 92, LDH 1085) on January 29, with a slow return to normal over the next six days. A lung "scan" on January 27 was interpreted as normal. That evening he complained of some vague upper abdominal discomfort, and during the next 48 hours he developed severe upper abdominal pain that required meperidine for relief. Mild abdominal distension developed, but his bowel sounds remained active and he had several loose bowel movements. There were no signs of peritoneal irritation.

On the morning of January 29 he had a brief episode of dyspnea, tachypnea and tachycardia.

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Apart from moderate bronchospasm, there was no change in physical findings. Chest film, electrocardiogram and a transvenous pulmonary angiogram showed no evidence of a pulmonary embolus. Later in the day the blood pressure dropped to 90-110/60-70 mm. Hg (average: 100/60) and remained at this level until February 1. The patient, who had been somewhat obtunded, now became disoriented but no other neurological deficits appeared. His oral temperature rose to 102-103° F. for several days. Blood cultures were negative. Random blood sugars at this time ranged from 100 to 150 mg. %. Repeated eosinophil counts (January 31 to February 10) varied from 120 to 400, with one count on February 7 as high as 500 per c.mm.; the leukocyte counts at this time varied from 6000 to 8000 per c.mm. On February 1 and 2, he again complained of severe pain, but now it seemed to be chiefly in the lower posterior thoracic region and again he required heavy analgesia. There was no local tenderness and no discolouration in the flanks, but there was some questionable fullness in the right loin. No change in the femoral or popliteal pulses was noted and no petechiae or splinter hemorrhages appeared on the legs. Repeated urinalyses were negative and many stools were also negative for occult blood. Marked abdominal distension, vomiting and absent bowel sounds developed. A plain film of the abdomen showed dilated loops of small bowel, and the psoas shadows could not be definitely identified. His hemoglobin, which was 15.7 g. per 100 ml. on admission, had fallen to 11.2 per 100 ml. on January 31 and further to 9.4 per 100 ml. on February 4. The platelet count varied from 170,000 to 300,000, and at no time was there any evidence of a generalized bleeding diathesis. An initial and only dose of phenprocoumon (Marcumar) 18 mg. was given on February 1 and the patient's prothrombin time on February 3 was 30 sec. (control 14 sec.). At this time, he was believed to be suffering from retroperitoneal hemorrhage and the heparin was stopped. The symptoms and signs gradually subsided over the next three days, by which time the thrombophlebitis was also clearing. On February 2 a major motor seizure occurred which was without localizing signs and left no residual neurological deficit.

On February 2 the serum electrolytes which had been normal on January 28 were now abnormal: sodium 120, potassium 4.9, chloride 82, and CO_2 combining power 25 mEq. per litre. During the following four days he received 150 mEq. of sodium daily. Serum sodium levels varied from 120 to 126 mEq. per litre while the 24-hour urinary sodium losses ranged from 60 to 80 mEq. of sodium. Serum osmolality on February 6 was 252 and the urine osmolality 315 mOm. per litre. With fluid restriction, the serum sodium returned to 132 to 134 mEq. per litre and there was concomitant clinical improvement. However, in the same period, the serum potassium had increased to 5.5 to 6.0 mEq. per litre. An intravenous pyelogram and renal scan showed normal-sized kidneys but with some suggestion of downward displacement of the right kidney. There was bilateral impairment of renal function. Endogenous creatinine clearance was 50 to 60 c.c. per minute, and blood urea nitrogen was 18 to 25 mg. %.

Evaluation of his adrenal function on February 10-11 showed 24-hour urinary 17-ketogenic steroids of 1.3 mg. per 24 hours (normal: 9 to 17 mg. per 24 hours) and 17-ketosteroids of 1.7 mg. per 24 hours (normal: 9 to 22 mg. per 24 hours). When this value was reported on February 20, adrenal insufficiency was considered likely and a four-hour intravenous adrenocorticotropic (ACTH) test was performed on February 21. Plasma cortisol determinations were made at 0, 2, 4 and 6 hours after starting a four-hour intravenous infusion of 75 units adrenocorticotropin (Acton X) in 40 ml. 5% glucose in water.[5] A low initial plasma cortisol level of 2.5 μg. per 100 ml. rising to a maximum at four hours of only 5.4 μg. per 100 ml. was reported the same day, and oral cortisone 37.5 mg. daily was begun after several intramuscular injections of cortisol hemisuccinate (Solucortef) had been given. This replacement therapy produced definite clinical improvement. The serum sodium rose from a low of 122-137 to 141 mEq. per litre, the serum potassium dropped from a high of 6.0 to 3.8-4.9 mEq. per litre, and the CO_2 combining power increased from a low of 21 mEq. per litre to 22-25 mEq. per litre. Since he seemed to have slight residual adrenocortical function, his response to ACTH stimulation was assessed again on March 18 and on April 21 to 25 to determine if any regeneration of adrenal tissue might have occurred. During these periods his steroid therapy was changed to dexamethasone 0.75 mg. daily because this drug is not measured in the assay. Since, by April 21, he had been receiving cortisone for two months, the intravenous test was done after the intramuscular administration of 120 units of ACTH daily for three days. On both these occasions the initial plasma cortisol levels were negligible and they did not rise during ACTH administration. In April, repeat renal function tests were within normal limits.

At time of writing the patient feels very well, has no specific complaints, and walks long distances daily. His dyspnea and cough have disappeared. His current medications include digitalis, cortisone acetate 37.5 mg. per day, and 9α-fluorohydrocortisone 0.1 mg. per day. On exposure to the sun he has developed a good tan.

DISCUSSION

The clinical course of patients who develop adrenal hemorrhage while on anticoagulants is not so characteristic as the fulminant course of the Waterhouse-Friderichsen syndrome where shock, hyperpyrexia, extensive purpura and hypoglycemia usually occur within 24 hours. The latter condition occurs mainly in children and young adults while the former is more common in persons 50 to 70 years of age and is generally associated with severe illness such as myocardial infarction or pulmonary embolism. Therefore,

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the primary disorder for which the anticoagulation is being given is frequently a ready explanation for most of the clinical features, and the correct diagnosis is not entertained, at least not early. According to Amador,[1] who recently reviewed the clinical features of 30 such cases, the hemorrhage occurred 2 to 13 days after anticoagulant therapy was begun, and there was no evidence of generalized bleeding. Contrary to the general view expressed by Cahill, Jenkins and Thorn,[6] excessive anticoagulant therapy did not appear to be responsible. Severe crampy upper abdominal pain with tenderness and guarding occurred most commonly, often accompanied by nausea and vomiting. Death occurred in two to eight days. Retroperitoneal hemorrhage was infrequent, mimicking an acute "surgical" abdomen. In most patients the adrenal hemorrhage was bilateral, but a few have been reported with unilateral hemorrhage who survived following operation.[7-9] Close analysis of cases of adrenal hemorrhage suggests that clinical acumen and perspicacity is necessary if these cases are to be diagnosed during life.[10, 11]

In this diagnosis, Amador believes that an eosinophil count above 50 per c.mm. is a most useful, although not specific, test. Certainly in the acute situation the collection and processing of 24-hour urine samples for 17-ketosteroids, 17-hydroxysteroids and/or 17-ketogenic steroids is much too slow. The determination of corticoids in random plasma samples is helpful but not fully adequate, since many factors can affect the plasma level. Indeed, the only satisfactory way of proving adrenal insufficiency is to demonstrate the inability to respond to ACTH. With newer methods of measuring steroids in plasma, a definitive diagnosis can be made within a few hours. Two rapid procedures for steroid determinations are available: fluorescence analysis and competitive protein-binding analysis. Their relative merits have been discussed recently.[12] In the case under discussion, once the possibility of adrenal insufficiency was seriously considered, the diagnosis was confirmed the next day using the competitive protein-binding method. In a seriously ill patient, dexamethasone and/or 9α-fluorohydrocortisone can be given during the ACTH test, since these steroids are not measured by this procedure. When such tests are not available immediately, the safest course is to treat the patient with steroids after first taking a blood specimen that is sent away for analysis at once. The diagnosis should be confirmed later by an ACTH test. If there is no adrenal insufficiency, no harm will have been done by the administration of high doses of steroids if these are not continued for more than two or three days.

The incidence of idiopathic adrenal hemorrhage in autopsy series varies from 0.14 to 9%,[13] but Thrash and Iri[14] believe that many cases are difficult to recognize at autopsy unless the gland is examined histologically; thus, the reported incidence in autopsy material may be too low. Botteri and Orell,[13] in reviewing the clinical features of 23 autopsied cases, noted that the clinical course was the same whether the hemorrhage was unilateral or bilateral with massive or multiple small areas of hemorrhage and infarction. They suggested that the adrenal hemorrhage may be due to medullary vein thrombosis, while multiple small areas of cortical bleeding and necrosis are caused by capillary damage of unknown etiology. Others have also described venous thrombosis but did not believe it was necessarily associated with hemorrhage and necrosis; in fact, it may be a result of it. Thrash and Iri also believe that ischemia due to arterial spasm or lowered blood pressure during shock would readily produce selective irreversible damage to the adrenal cortex because it has such a high rate of metabolism—up to three times that of liver or kidney, for example—and is supplied by some 50 tiny arteries although it is drained by only one vein. Greendyke[15] reviewed 33 cases of "idiopathic" adrenal hemorrhage and compared them with 22 cases of fatal meningococcemia. He noted the similarity in histology in the two series, including the "tubular degeneration" of the adrenal cortex described by Rich[16] and Selye and Stone.[17] Such changes have been produced by the administration of very large doses of ACTH, and several cases of adrenal hemorrhage have occurred in patients receiving ACTH.[18-20]

Some experimental work has been done to elucidate the underlying mechanism in adrenal hemorrhage. Wilbur and Rich[21] were able to produce adrenal hemorrhages in rats by ACTH injections and this has been confirmed by many others. On the other hand, hypophysectomized animals subjected to stress did not develop such hemorrhages.[22] Barta, Beregi and Jellinek[23] produced similar lesions in rabbits by treating them with cortisone and then inducing insulin shock. Levin and Cluff[24] could regularly sensitize the adrenal glands to hemorrhage by pretreating their animals with Thorotrast, ACTH, or endotoxin—all of which will produce intense adrenal stimulation—and then, after several hours, injecting endotoxin. Thorotrast alone was also effective. In addition to adrenal stimulation, Mosley and Cluff[25] showed that this agent produced multiple coagulation defects and vascular wall injury. Suppressive doses of steroids given before these agents were ad-

ministered resulted in a "marked suppression in the frequency and severity of the adrenal hemorrhage", whereas steroids given just before the endotoxin injection did not produce so great a suppression. This is in keeping with the fact that corticosteroids may, under certain circumstances, suppress or enhance the noxious effects of endotoxin shock, depending upon the time relationship of its administration. Pretreatment of the animals with heparin did not prevent the adrenal hemorrhage. Elaboration of adrenergic vasoactive substances by the adrenals did not seem to play a significant role in the pathogenesis. In a well-controlled experiment, Hoelscher[26] showed that rabbits stressed by immersion of a hind foot in a freezing solution developed degenerative changes in the adrenal cortex. Treatment with heparin resulted in more severe degenerative changes and a higher mortality. Margaretten *et al.*[27] produced adrenal hemorrhage in rats by disseminated intravascular clotting and inhibition of fibrinolysis. If this were the etiological basis of the Waterhouse-Friderichsen syndrome, the steroids frequently administered as treatment may indeed be deleterious because they may potentiate the Shwartzman reaction.

Heparin is known to affect adrenal function in another way. Wilson and Goetz[28] described a patient in whom heparin produced selective hypoaldosteronism; the work of Conn *et al.*,[29] showing that heparin inhibits aldosterone synthesis in man, supports this. However, this action does not seem to bear any relationship to the occurrence of adrenal hemorrhage.

The following train of events probably provides a basis for spontaneous adrenal hemorrhage. Stress produced in a variety of ways causes release of adrenocorticotrophin, which produces intense adrenal stimulation, especially of the midcortical region or zone fasciculata with a marked increase in its metabolism. This results in cortical lipid vacuolization and focal cytolysis with formation of "lumina" or "tubular degeneration". Under these conditions, the cortex is more susceptible to injury. Involvement of vascular endothelium, due to as yet unidentified factors, follows, with the production of hemorrhage and infarction, although death due to adrenal failure can occur without gross hemorrhage or infarction. Agents such as anticoagulants tend to extend any injury.

In the present case one may speculate that the patient had a myocardial infarction on January 25-26, with increased production of ACTH over the next few days. Bilateral adrenal hemorrhage with retroperitoneal hematoma occurred on January 27-29, with a further extension on February 1-2. A minimal degree of function, possibly mainly in the glomerulosa, was maintained for several weeks, but the final result was probably complete destruction of both glands.

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